

John Doe

[johndoe@gmail.com](mailto: johndoe@gmail.com) | [linkedin.com/in/johndoe](https://www.linkedin.com/in/johndoe) | github.com/johndoe | [xxx-xxx-xxxx](#)

EDUCATION

University of Connecticut Storrs, CT
Bachelor of Science in Computer Science, Minor in Mathematics May 2027

- Related Coursework:** Computer Architecture, Linear Algebra, Systems Programming, Advanced Algorithms, Software Engineering, C++ Programming, Data Structures & Algorithms, Discrete Mathematics, Cryptography

EXPERIENCE

Software Engineer Intern Jun 2026 - Aug 2026
Pure Storage Santa Clara, CA

- Incoming Summer 2026, C++

Quantitative Developer Jan 2026 - Present
Husky Quantitative Group - University of Connecticut Storrs, CT

- Engineered core execution engine for a student run hedge-fund, enabling quantitative research and trading
- Wrote C++ engine with order lifecycles, partial fills, and position updates, reducing backtest runtime by 87%
- Implemented deterministic backtesting engine with market data replay and exchange adapters using **Python**
- Designed strategy validation engine through AST-based static analysis and containerized execution with **Docker**

Undergraduate Research Assistant Aug 2025 - Present
Machine Intelligence and Data Science (MINDS) Lab - University of Connecticut Storrs, CT

- Co-authored paper researching Graph Neural Network robustness, focusing on reliability under environmental shifts
- Implemented a **Mixture-of-Experts** encoder into Graph Foundational Model for hybrid-routing using **PyTorch**
- Increased output stability by 14% after fine-tuning Graph Neural Networks with a **Lipschitz**-regularized classifier

Software Engineer Intern Sep 2025 - Nov 2025
Iota -1 LLC - University of Connecticut Storrs, CT

- Created and published **CodeShuffler**, an open-source tool for generating paper-based programming assessments
- Designed a **Python + PyQt** interface with syntax highlighting, PDF + HTML previewing, and shuffled exam output

Software Engineer Intern Jun 2025 - Aug 2025
Modo Labs Boston, MA

- Led the development and adoption of AI knowledge agent into product offerings using FastAPI, Python + LangChain
- Engineered a **Retrieval-Augmented Generation (RAG)** pipeline using a **Pinecone** vector database along with **Redis**-backed prompt caching across user sessions, achieving a response accuracy of 97% and reducing API costs

PROJECTS

Low-Level Sound Synthesizer | C++ 20, Windows API, Multithreading, Audio Programming

- Built a sound synthesizer in C++ leveraging **Windows API** for real-time audio with sub-millisecond latency
- Designed multithreaded, event-driven sound wave synthesis with phase continuity and low harmonic distortion

Relational Database Kernel | C++ 20, Multithreading, Storage Systems

- Designed a single-node database kernel involving page-based storage, buffer pool management, and query execution
- Engineered a lock manager to coordinate buffer pool pages and **B+ tree nodes**, resolving deadlock risk via ordered lock acquisition and sustaining concurrent insert, delete, and scan workloads across multiple worker threads

PUBLICATIONS

- CodeShuffler v2: A System for Paper-Based Coding Assessments in the Era of Generative AI:** Software system for creating and distributing paper-based coding assessments | J. Doe, X. X | ASEE '26 (1st Author; pending)
- Generalizing GNNs with Tokenized Mixture of Experts:** Framework for improving GNN task-classification robustness under environmental shifts during deployment | X. X, J. Doe, et al. | KDD '26 (Co-Author; pending)

TECHNICAL SKILLS

Languages: Python, C++, C, TypeScript, SQL, Bash
Frameworks & Libraries: Flask, FastAPI, PyTorch, PyQt
Tools: CMake, Ninja, AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux